

Hello everyone! Next, I'm going to talk about three landmark cryptocurrency security incidents. And we are trying to find their shared risk factors, and then analyze the ecosystem challenges.

1. So let's have a Briefly introduction of the Three Incidents:

Mt. Gox:

This company was latched in 2010. And it once handling over 70% of all bitcoin transactions over the world by early 2014. So it was really a huge company.

But just at the end of 2014. Hackers gained access to the private keys because their hot wallets were not secured properly. And as a result, about **7hunderd and fourty** thousands BTC were stolen.

Poly Network:

What is poly network? It is kind of cross-chain protocol which is very popular that time. It allows you move 'assets between different blockchains. For example: You can exchange your bitcoins for other virtual currencies like Ether.

This incident happened in August 2021, an attacker found a validation bug in its smart-contract logic. And He soon stole about **6hunderd and ten** million **dollars** worth of tokens. Although it was eventually returned, it still had a significant negative impact

Ronin Network:

Ronin Network is similar to the Poly example but was just built specifically for a game named Axie Infinity. This Network enables players to make high-volume, low-fee ETH and USDC transfers.

This incident happened in March 2022, a hackers Group compromised the private keys of 2 out of its 9 nodes and stole about **6hunderd and fifteen** million **dollars**

So Now that we've reviewed those three incidents . We can find that all three hacks had the same 4 risks:

- ❓ Hot wallets and private keys were not properly secured, so attackers could grab keys and drain funds.
- ❓ About the **Smart-contract code** and **cross-chain checks**. Many companies ignore this because its management reasons.
- ❓ The Critical operations depended on just a few keys or nodes. For example, in the third incident, there were actually only two private keys were compromised, but the whole system was collapsed.
- ❓ There was no real-time monitoring or rapid response, so all the thefts went on for days before anyone noticed."

Knowing this I'm going to make a summary of the ecosystem challenges:

First, immature security practices:

For example, there is no industry-wide standards for private-key, multi-signature policies, protocols and so on. When there is no clear rules for handling private keys, they can easily get leaked or lost

The Second one is similar: There is no unified vulnerability-disclosure channels or formal certification bodies. So bugs are more likely to be exposed.

Third challenge is about decentralization:

On paper, the protocol claim to be decentralized, but real-world deployments still rely on a handful of trusted nodes or key-holders.

The last one is the lack of clear legal frameworks or investor-protection mechanisms.

These ecosystem challenges demonstrate why use only technical fixes are not enough—we need coordinated standards, rules, oversight to truly secure the space.